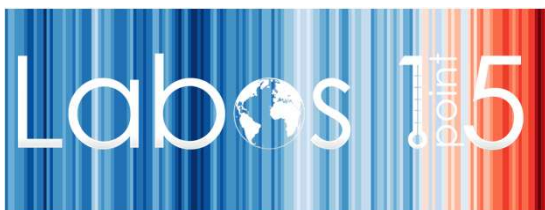


The GHG footprint of an experimental laboratory



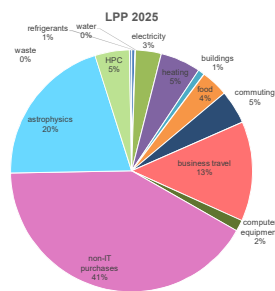
Laurence Rezeau, Fouad Sahraoui
with support from the LPP's Administration team and the "Climat" Working Group

Laboratoire de Physique des Plasmas



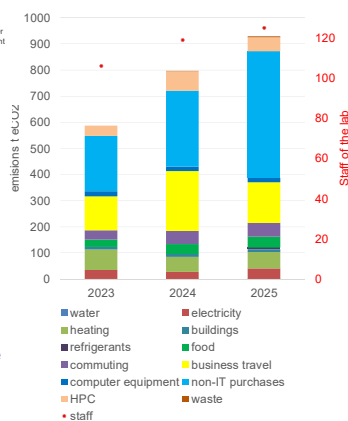
- **Measure: GES 1point5** gives the tool to estimate a carbon footprint and build the greenhouse gas (GHG) inventory of our structure. This in order to bring food for thought on the levers for action to reduce GHG emissions.
- **Share: Transition 1point5** enables laboratories to describe the actions they are taking to sustainably reduce their environmental footprint. It also enables observers to explore these actions for inspiration.

The GHG footprint of LPP in 2025



Footprints of Tokamaks
And Regional computing centers
are missing

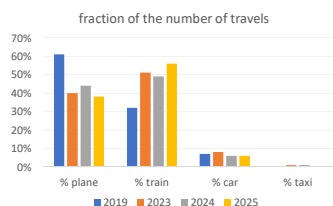
Comparison excluding astrophysics



- The 2025 footprint corresponds to around 9 t eCO2/person including astrophysics or 7 t without
- The « astrophysics » footprint is highly variable from a year to another
- The global footprint has increased over 3 years (58%), more than the staff (18%)

What have we done?

- Individual decision of changing means of transport for commuting (car to bike) or business travel (plane to train when possible)
- Lab decision of paying 1st class for long train travels (if > 4h)

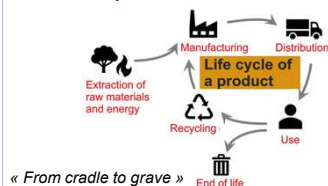


The footprint of plane travels is 97.4% of all the travels footprint

- Encourage the sharing of equipments through a numerical platform

How are the GHG emissions computed?

- For a particular item, a Life Cycle Assessment is performed



It summarizes the emissions and gives an emission factor which is related to the price : on average 0.35 kg eCO2/€

All GHG are converted in equivalent CO2

- For the lab purchases we use codes (NACRES) that classify the nature of the purchase and give the emission factor (using data bases from ADEME, CEDA and EPA)

Examples:

Very pure Argon (code GA11, 1.9 kg eCO2/€)

Preliminary mechanical analyses for PO iEPC (code RD01, 0.14 kg eCO2/€)

Analyzing the accounts gives the footprint

- For a space mission after launch, the process is the same as for any other item

Knödseder et al, 2022

$$F_{\text{construction}} = 48 \text{ t eCO}_2 \text{ M(kg)}$$

$$F_{\text{operation}} = 1 \text{ t eCO}_2 \text{ M(kg)/year}$$

Where M is the lift-off mass

But the question afterwards is:

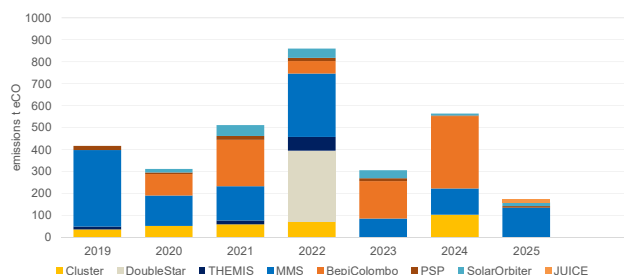
How is the footprint displayed over the labs participating in the mission ?

Knödseder et al, 2022

- Amortization of the construction footprint over 18 years
- $F_{\text{mission/lab}} = F_{\text{infra}} \frac{\text{nb of coauthors citing mission in the lab}}{\text{nb of coauthors citing mission in the world}}$
- Using ADS database

What are the difficulties?

- No real measurement of the fluid consumption or waste production; the footprint is supposed to be proportional to the surface of the lab (or to the staff)
- No control over the insulation of buildings or the heating system
- Palaiseau campus not easy to reach by public transport
- For space missions, external constraints come from the agencies (companies from which we must buy, international meetings, ...) and purchases are very expensive (with respect to lab equipment)
- The emissions attributed to space missions are highly variable, more than our real activity : the « thermometer » is not trustable



Another way of displaying the footprint over the labs could be to take into account the PI-ship or Co-I-ship on experiments

